

Data Mining: For Gold



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You are moving your shopping cart towards the billing counter.

The barcode of each product in your cart has already been automatically transmitted one by one to one of the billing computer systems. This is your first visit to this department store. You are allotted a loyalty card that has a customer ID number and are asked to fill a short form that asks details like your age, gender, income and occupation. A few days later you receive an email from the store informing you about some good offers on headphones; you had purchased an iPod among other things during your last visit. This is data mining at work.

MCI Communications Corp., the long-distance carrier, sifts through almost one trillion bytes of customer phoning data to develop new discount calling plans for its different types of customers. Land's End, a retail chain, identifies which of its two million odd customers can be sent exclusive mailers on certain specific clothing items that will go well with their existing wardrobe.

Data warehousing and data mining

Data warehousing is all about capturing data on a periodic basis from various sources and storing it on computers in an organised manner, that is, in virtual warehouses.

Fig. 1: Basic steps involved in the process of data mining

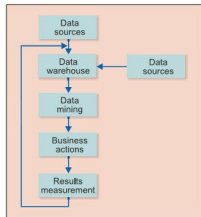


Fig. 2: A bitcoin mining hardware set-up that uses custom chips known as ASICs to focus their processing power on bitcoin algorithms (Source: www.datacenterknowledge.com)

Let us take its application in marketing to customers as an illustration. Some common sources of data are details of purchases made at retail outlets, marketing research interviews with customers or simply day-to-day customer transactions (such as amount of money deposited or withdrawn from a bank). Usually, the starting point is a customer database that needs to be elaborate in terms of variables such as demographics like age, family size and income. Customers leave traces of their purchasing behaviour in store scanning data, catalogue purchase records and customer databases.

Customers' actual purchases reflect their true and actual preferences, and could often be more reliable than answers given during market research surveys. Data mining refers to the analysis of this raw data, which is collected to generate useful and meaningful information, such as finding patterns in terms of products purchased most often and building customer profiles in terms of low, medium and heavy buyers. This information can then be used for business actions like special promotions and loyalty programmes.

A brief history

Data mining was born in the minds of professors and doctorates who used their expertise in statistics to develop algorithms

that form the foundation of data mining today. In the initial days, these algorithms were custom-coded using programming languages such as FORTRAN.

Early applications of data mining were more academic in nature. As this work evolved, different vendors fabricated new levels of tooling to make application of these data mining algorithms more user-friendly.